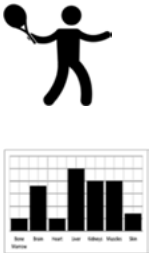
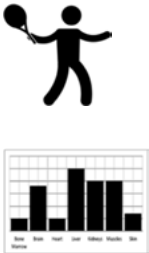


1.1.e. Effects of exercise on body systems

Learners will develop their knowledge and understanding of the short and long-term effects of exercise on muscles and bones, the heart and the respiratory system. They will be able to apply understanding of these effects to examples from a range of physical activities and sports.

Learners will be able to collect and use data in this section related to both short-term and long-term effects of exercise.

Topic area	Learners must:
Short-term effects of exercise 	- understand the short-term effects of exercise on: - muscle temperature - heart rate, stroke volume, cardiac output - redistribution of blood flow during exercise - respiratory rate, tidal volume, minute ventilation - oxygen to the working muscles - lactic acid production - be able to apply the effects to examples from physical activity/sport - be able to collect and use data relating to short-term effects of exercise.
Long-term (training) effects of exercise 	- understand the long-term effects of exercise on: - bone density - hypertrophy of muscle - muscular strength - muscular endurance - resistance to fatigue - hypertrophy of the heart - resting heart rate and resting stroke volume - cardiac output - rate of recovery - aerobic capacity - respiratory muscles - tidal volume and minute volume during exercise - capillarisation - be able to apply the effects to examples from physical activity/sport - be able to collect and use data relating to long-term effects of exercise.